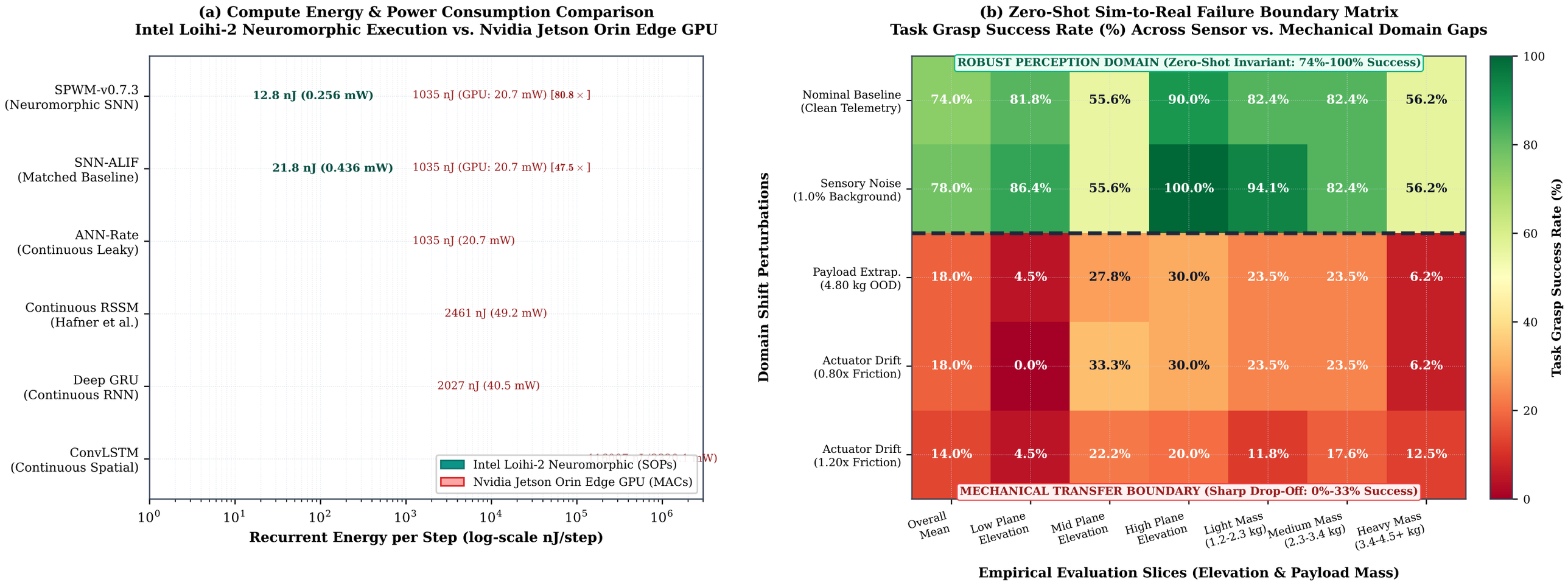

(a) Compute Energy & Power Consumption Comparison
Intel Loihi-2 Neuromorphic Execution vs. Nvidia Jetson Orin Edge GPU
SPWM-v0.7.3 (Neuromorphic SNN)
SNN-ALIF (Matched Baseline)
ANN-Rate (Continuous Leaky)
Continuous RSSM (Hafner et al.)
Deep GRU (Continuous RNN)
ConvLSTM (Continuous Spatial)
12.8 nJ (0.256 mW)
1035 nJ (GPU: 20.7 mW) [80.8 ×]
21.8 nJ (0.436 mW)
1035 nJ (GPU: 20.7 mW) [47.5 ×]
1035 nJ (20.7 mW)
2461 nJ (49.2 mW)
2027 nJ (40.5 mW)
Intel Loihi-2 Neuromorphic (SOPs)
Nvidia Jetson Orin Edge GPU (MACs)
10^0 10^1 10^2 10^3 10^4 10^5 10^6
Recurrent Energy per Step (log-scale nJ/step)
(b) Zero-Shot Sim-to-Real Failure Boundary Matrix
Task Grasp Success Rate (%) Across Sensor vs. Mechanical Domain Gaps
Domain Shift Perturbations
ROBUST PERCEPTION DOMAIN (Zero-Shot Invariant: 74%-100% Success)
Nominal Baseline (Clean Telemetry)
74.0% 81.8% 55.6% 90.0% 82.4% 82.4% 56.2%
Sensory Noise (1.0% Background)
78.0% 86.4% 55.6% 100.0% 94.1% 82.4% 56.2%
Payload Extrap. (4.80 kg OOD)
18.0% 4.5% 27.8% 30.0% 23.5% 23.5% 6.2%
Actuator Drift (0.80x Friction)
18.0% 0.0% 33.3% 30.0% 23.5% 23.5% 6.2%
Actuator Drift (1.20x Friction)
14.0% 4.5% 22.2% 20.0% 11.8% 17.6% 12.5%
MECHANICAL TRANSFER BOUNDARY (Sharp Drop-Off: 0%-33% Success)
Overall Mean
Low Plane Elevation
Mid Plane Elevation
High Plane Elevation
Light Mass (1.2-2.3 kg)
Medium Mass (2.3-3.4 kg)
Heavy Mass (3.4-4.5+ kg)
Empirical Evaluation Slices (Elevation & Payload Mass)
Task Grasp Success Rate (%)
100
80
60
40
20
0